

Artificial Intelligence & Information Economics

- IT-250: Digital Economy | Unit 5
- Lecture 4 – AI & Information Economics
- Example: Using AI to recommend products like Netflix shows or Daraz shopping suggestions.

Link to Previous Lecture

- We learned about Asymmetric Information
- Problems: Adverse Selection & Moral Hazard
- Today: How AI helps manage and analyze information

Example: AI can reduce the hidden information problem in online markets in Nepal.

What is AI? (Kid-Friendly)

- AI = smart computer programs
- Can think, predict, decide like humans
- Examples: Google Maps predicts traffic; YouTube recommends videos
- Nepal Example: SajiloPay app suggests services based on past usage.

AI in Economics

- Processes huge data quickly
- Helps firms & consumers make better decisions
- Reduces cost of gathering/analyzing information

Example: Banks in Nepal use AI to analyze loan applications quickly.

AI as a Strategic Tool

- Companies gain competitive advantage
- Helps answer: what customers will buy, products to stock, dynamic pricing
- Nepal Example: eSewa predicts popular services; Daraz suggests trending products
- Kid Example: Candy shopkeeper predicts which candies sell fastest.

AI-Driven Personalization

- AI learns user habits
- Shows recommendations tailored to you
- Kid Example: Netflix shows cartoons you like
- Nepal Example: YouTube Nepal recommends local news and music
- Real Scenario: Daraz recommends shoes based on previous purchases.

Recommendation Engines

- Predicts what you like based on past behavior
- Used in e-commerce, media, social platforms
- Nepal Example: Foodmandu suggests restaurants you might like
- Kid Example: Spotify recommends songs you often listen to.

How AI Changes Information Economics

- Reduces search cost → find info faster
- Lowers processing cost → easier comparisons
- Makes info more accessible
- Nepal Scenario: Students can find scholarship info online quickly using AI-powered portals.

AI & Decision-Making

- Helps consumers make smart choices
- Helps firms predict demand & optimize supply
- Reduces errors from limited knowledge
- Nepal Example: Banks use AI to decide loan approvals; Health apps predict local disease outbreaks
- Kid Analogy: AI chooses the best path in a maze for you.

AI & Markets

- Makes markets more transparent
- Reduces asymmetric information
- Detects fraud or fake sellers
- Nepal Example: Verified reviews on Daraz; AI detects banking fraud
- Scenario: Daraz flags fake seller profiles automatically.

AI & Strategies

- Customize pricing for users
- Dynamic pricing using AI
- Identify trends before competitors
- Nepal Example: Tourism packages for peak season; Airlines adjust ticket prices dynamically
- Kid Example: Lemonade stand adjusts prices if more kids show up.

AI and the 'New Economics of Information'

- Old: Information costly & slow
- With AI: cheap, fast, personalized
- Makes markets efficient and responsive
- Nepal Example: e-commerce platforms provide real-time pricing and product recommendations.

Impacts on Consumers

- Better product recommendations
- Smarter buying decisions
- Personalized offers save time & money
- Nepal Example: Foodmandu app suggests meals based on dietary preferences
- Kid Example: Toy store tells which toys you'll love.

Impacts on Businesses

- Optimize marketing & supply chain
- Predict demand & customer behavior
- Reduce waste & costs
- Nepal Example: Daraz predicts peak buying during Dashain to stock products
- Scenario: Coffee shop adjusts inventory using AI predictions.

Impacts on Labor Markets

- AI replaces repetitive tasks
- Creates demand for AI-skilled workers
- Nepal Example: AI in fintech, data analytics, e-commerce
- Kid Analogy: Robot does repetitive chores, humans do creative work.

Ethical Considerations

- Bias in AI predictions
- Privacy concerns
- Transparency & explainability required
- Nepal Example: Finance apps and government AI projects must protect data
- Scenario: Avoid unfair loan denial due to biased AI.

Kid-Friendly Analogy: Smart Robot Shopkeeper

- Knows your favorite candies 🍬
- Tracks last purchases
- Suggests toys you might like
- Predicts which items sell fast
- Nepal Scenario: Daraz or Foodmandu acts like robot shopkeeper
- Lesson: AI reduces search time and predicts choices.

Deep Analogy: Smart Librarian

- Huge library with millions of books
- Without AI: search shelf by shelf → hours wasted
- With AI: smart librarian finds books instantly
- Nepal Scenario: Students searching for scholarships online
- Kid Analogy: AI is a helpful librarian that knows every book.

AI in Everyday Life Scenarios

Scenario 1: Grocery Shopping

- Daraz/SastoDeal tracks past purchases
- Suggests products, predicts peak demand

Scenario 2: Health & Fitness Apps

- Predicts activity levels, suggests workouts/diets
- Lesson: AI acts as a personal assistant
- Nepal Example: Seasonal grocery price predictions using AI.

Key Exam Terms

- Artificial Intelligence (AI)
- Recommendation Engine
- Personalization
- Search Cost
- Information Accessibility
- Market Efficiency
- Example Terms in Nepal Context: Daraz recommendations, eSewa prediction, fintech AI.

Simple Summary

- AI processes, analyzes, predicts information
- Reduces cost & effort of getting information
- Makes markets transparent, efficient, and smarter
- Impacts consumers, businesses, and labor markets
- Nepal Example: Daraz, eSewa, online scholarship portals.

Discussion Questions

- How does AI make buying online easier in Nepal?
- Can AI remove all asymmetric information?
- Should AI decisions always be trusted?
- Scenario: If AI recommends a product and it is poor quality, who is responsible?

Discussion Questions

1 How does AI make buying online easier in Nepal? NP

Answer:

AI simplifies online shopping by:

- **Recommending products** based on your past purchases
 - Example: Daraz suggests shoes or clothes you often buy
- **Predicting trends** so sellers stock popular items
 - Example: Daraz or SastoDeal predicts high demand for rice or snacks during Dashain
- **Reducing search effort:** AI filters thousands of products to show the best matches
- **Providing personalized deals:** Apps like Foodmandu show meals you may like

Analogy:

AI acts like a friendly shopkeeper who already knows what you want, so you don't have to search everywhere.

Discussion Questions

Can AI remove all asymmetric information? ✖

Answer:

- **No, AI cannot remove all asymmetric information.**
- It reduces **information gaps**, but some limits remain:
 - Fake reviews or biased data
 - Sellers hiding real product quality
 - Algorithm errors or incomplete data
- AI helps, but **human judgment is still needed**

Nepal Example:

- Even if Daraz shows 5-star shoes, the seller may send low-quality ones.
- AI improves transparency, but **can't guarantee perfect info.**

Analogy:

AI is like a smart librarian, but if a book is torn and not in the database, AI can't fix that.

Discussion Questions

Should AI decisions always be trusted? 🤖

Answer:

- **Not always.** AI is a tool, not a human.
- Reasons to be cautious:
 - Data bias → unfair or wrong recommendations
 - Limited understanding of local context
 - AI doesn't understand "ethics" or emotions
- Best approach: **use AI as guidance, but verify before acting**

Nepal Example:

- Banks use AI for loans, but manual checks ensure risky borrowers aren't unfairly approved/denied
- E-commerce ratings should be cross-checked with multiple sources

Analogy:

AI is like a helpful robot. It can guide you, but sometimes you still need your own eyes to check.

Discussion Questions

Scenario: If AI recommends a product and it is poor quality, who is responsible? ⚖️

Answer:

- Responsibility depends on **platform rules & seller accountability**:
 - **Seller**: Should provide accurate product description
 - **Platform**: Should monitor fake sellers and AI recommendations
 - **Consumer**: Should read reviews and check return policies
- AI only **guides**; it is **not legally liable** in most cases

Nepal Example:

- You buy a phone on Daraz based on AI recommendation, but it's faulty:
 - Daraz customer service allows return/exchange
 - Seller is responsible for misrepresentation
 - AI was only the recommendation tool

Kid analogy:

The robot suggests a toy, but if the toy breaks, the shopkeeper is responsible, not the robot.